

Implementation and Feasibility Plan

Samadhan (समाधान): from a first pilot to all of Madhya Pradesh

1. Feasibility today (what is running)

The prototype is built and tested: a FastAPI core, a website with voice and GPS, a Supabase database with a private audio bucket, and a Streamlit officer dashboard. It runs four departments (water, electricity, roads, sanitation) and a Human Evaluation desk on one engine. Everything used is on free tiers, and everything server-side is stateless except the database.

2. Phased plan

Phase	Goal	Work	Depends on	Exit check
0. Prototype (done)	Prove the flow	Website chat, voice, routing, tickets, dashboard, status	none	Ten scenarios exercised; 415 + 35 tests pass
1. Make the data real	Replace demo data	Verified ward and office data for the first pilot districts, starting with Bhopal (RTI or a visit to the local body); ward centre points from the ward boundary file; officer roles; resolve BMC-versus-PHED ownership	BMC cooperation	GPS picks the correct ward; no (DEMO) rows in the pilot
2. Harden for real users	Safe to open to the public	Real SMS OTP replacing today's demo PIN; rate limits on /message and /speak ; per-department officer accounts; officer notification when a ticket arrives; retention policy and consent notice	Phase 1	Abuse test passes; officers see only their department
3. First pilot districts	Measure real impact	live pilot with a few wards, then whole districts; track reassignments, time to first status change and voice use	Phase 2	The four impact metrics in item 05 have real numbers
4. Add departments and districts	Scale across MP	One YAML service spec plus office rows per department; a central department registry file with checks; another city as a second pilot	Phase 3	A new department goes live with no engine code change
5. Integrate	Fit the existing ecosystem	Ticket hand-off to MPOnline's Customer Solutions Hub / CM Helpline; WhatsApp channel on the same engine; DigiLocker-based document help with citizen consent	MPOnline agreement	Tickets land in the existing system

Phase order matters: phone verification and rate limiting come before any public launch, because an open SMS endpoint costs money per message and invites fake complaints.

3. Resources and cost (indicative, to be confirmed)

Item	Today	At pilot scale
LLM (Groq free tier, Gemini fallback)	Free; Groq free tier is about 8,000 tokens per minute per model, so bursts can exhaust it	Paid tier
Speech (Sarvam)	Free tier	Paid, per use

Item	Today	At pilot scale
Database and storage (Supabase)	Free; can pause after 7 days of inactivity	Paid plan
Hosting (Render)	Free; free hosts sleep when idle	Paid plan
SMS OTP for phone verification	Not in the product; planned after the SMS licence is obtained	Per-message cost; Indian SMS needs DLT (sender and template) registration, which takes days
Exact figures	Not yet quoted	To be confirmed with real price quotes

4. Risks and how we handle them

Risk	Mitigation
AI misroutes a complaint	Validator against the spec; confidence threshold; <code>needs_review</code> ; officer reassign with a logged correction
Provider outage or rate limit	LLM chain of four models; ASR fallback; short timeouts; Hindi error messages
Spam or fake complaints	Phase 2: phone verification, rate limits, duplicate detection
Wrong or unverified office data	Verified-or-labelled rule: every unverified row is marked DEMO; a registry with source and verified fields is proposed
Privacy and DPDP Act compliance	Minimal data by design; a full review (consent, retention, grievance officer) is a Phase 2 task, and we do not claim compliance today
Free tier limits	Move to paid plans before any public pilot

5. Team and process

A two-person team, working spec-first: every module has a written spec (`docs/specs/S01` to `S31`) and plan before code. The repository holds the specs, tickets and research so a new developer can pick it up (`docs/ONBOARDING.md`).